Making Salts

Aim: to make and name salts by neutralisation reaction.

Equipment:

- Sodium hydroxide solution
- Beaker
- 50 cm^3 measuring cylinder
- Hydrochloric acid
- Dropping pipette
- Stirring rod or capillary tube
- Universal indicator paper
- Evaporating basin
- Bunsen burner
- Gauze
- Tripod
- Heat proof mat

Safety Information

Acid – corrosive to skin and eyes

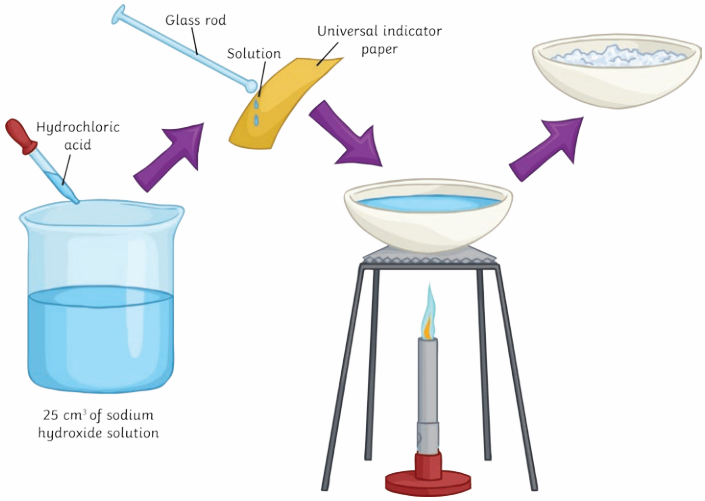
Wear goggles, clean spills immediately.

Glassware – risk of cuts if broken.

Apply pressure to wounds and inform teacher immediately.

Bunsen burner – risk of burns.

Place under cold running water for ten minutes. Inform teacher.



Method (series circuit):

Step 1: **You must wear goggles at all times**. Measure 25cm3 of sodium hydroxide solution into a beaker.

Step 2: Drop by drop, add hydrochloric acid to the sodium hydroxide.

Step 3: After every few drops, take a small sample on the stirring rod or capillary tube and test the pH. Keep adding acid until you have obtained a neutral pH result (pH 7 – green).

Step 4: Once you have a neutral solution, pour it into an evaporating basin.

Step 5: Using a tripod, mat, gauze and Bunsen burner set-up as shown above, heat the solution, until about 1/3 remains. This is a saturated solution.

Step 6: Once the equipment is cool, move the evaporating basin to a warm place in the room, and leave to evaporate completely overnight.

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Questions

1. What is meant by neutralisation?
2. What is a saturated solution?
3. What is the name of the salt produced in the reaction?
4. Write a word equation for the neutralisation reaction in your investigation.
5. Name the salt produced in these reaction:
   a. Hydrochloric acid and potassium hydroxide
   b. Sulphuric acid and calcium carbonate
   c. Nitric acid and calcium hydroxide

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Answers

1. What is meant by neutralisation?
   An acid and a alkali react together and produce a salt and water with neutral pH.

2. What is a saturated solution?
   A saturated solution cannot dissolve anymore solute. In this example no more salt can be dissolved.

3. What is the name of the salt produced in the reaction?
   Sodium chloride

4. Write a word equation for the neutralisation reaction in your investigation.
   hydrochloric acid + sodium hydroxide → sodium chloride + water

5. Name the salt produced in these reaction:
   a. Hydrochloric acid and potassium hydroxide
   Potassium chloride
   b. Sulphuric acid and calcium carbonate
   Calcium sulphate
   c. Nitric acid and calcium hydroxide
   Calcium nitrate